METHOD 22 / A COMPLETE BEGINNER GUIDE

Wavelet correlation and coherence

Question this method answers: At which time–scale (horizon) are two FX return series linked, and when during the sample does that linkage appear?

Wavelets localize fluctuations in both time t and scale s. Correlation compares coefficients at a scale; coherence adds a smoothed strength $R^2(s,t)$ with separate phase.

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Teaching returns only — no published wavelet coherence maps from snapshots.

Prepared for: 22 - Wavelet Correlation and Coherence. README status: not started.

02 / THE BASIC IDEA

See both time and horizon

Rolling Pearson r slides one time window. Wavelets use many scales s at once: short scales react to fast wiggles; long scales react to slow drifts. You can compare EUR/USD and GBP/USD at multi-week horizons while still seeing when linkage fades.

Two related objects

ρ_wavelet(s)	Correlation of wavelet coefficients at scale s across time.
$R^2(s,t)$	Local coherence strength after smoothing cross and auto spectra.
Phase	Timing offset — separate from R^2 magnitude.

03 / RETURNS ON EACH DAY

FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

1.1000 → 1.1110 is about +1% simple return. Feed return series into the transform — not raw price levels — unless you have a documented reason.

Example from the formula sheet: two majors may correlate on multi-week scales while short horizons look weak during the same calendar span.

04 / W(S,T)

Wavelet coefficients

Continuous wavelet transform (CWT) produces complex coefficients W_x(s,t) for each scale s and time index t. Real-part correlation is a teaching simplification; full coherence uses complex conjugates in the cross term.

Mother wavelet choice (Morlet, Paul, etc.) changes scale meaning — freeze the wavelet name in experiment NOTES.

05 / GRINSTED, MOORE AND JEVREJEVA (2004)

Research notes

Application of the cross wavelet transform and wavelet coherence to geophysical time series. Nonlinear Processes in Geophysics, 11, 561–566.

Popular reference for wavelet coherence diagrams: arrows for phase, contour lines for significant regions. FX researchers reuse the workflow with return series instead of climate variables.

Significance testing needs careful handling of autocorrelation — do not copy climate defaults blindly to FX.

06 / SMALL MATH DICTIONARY

Symbol dictionary

Symbol	Meaning
W_x(s,t)	Wavelet coefficient of series x at scale s, time t.
s	Scale / effective horizon (depends on wavelet and sampling).
ρ_wavelet(s)	Corr(W_x(s,·), W_y(s,·)) on real parts or as defined.
$R^2(s,t)$	Smoothed local coherence strength.
S	Smoothing operator in time and/or scale.
*	Complex conjugate in cross-wavelet spectrum.

Section 22 in forex-formulas-explained.pdf lists the same ρ_wavelet and R^2 structure.

07 / SCALE-BY-SCALE SCORE

Wavelet correlation

```
rho_wavelet(s) = Corr( W_x(s,·), W_y(s,·) )
```

Compute one correlation per scale using coefficient series over t. Compare scales side by side — a high ρ at large s with low ρ at small s matches the “linked on weeks, not days” story.

```
Teaching: Morlet CWT, scale=2.0, rho_wavelet ≈ 0.9991
```

Demean coefficients before correlating to match Pearson-style centering on the coefficient path.

08 / LOCAL STRENGTH

Wavelet coherence R²

```
R^2(s,t) = | S( s^-1 W_x W_y* ) |^2 /
( S(s^-1 |W_x|^2) * S(s^-1 |W_y|^2) )
```

Smoothing S reduces noisy speckles in raw cross-wavelet power. $R^2(s,t)$ lies in [0, 1] like spectral coherence but is indexed by both scale and time.

Phase arg(W_xy) shows lead/lag locally — coherence magnitude alone does not show direction or causation.

09 / TWO-SIDED FILTERS

Edge leakage warning

CWT uses filters that extend beyond each t. At series starts and ends, coefficients borrow from missing neighbors — edge regions are unreliable (cone of influence in Grinsted plots).

Two-sided non-causal filters must never use future returns in live forecasting — research on historical panels is fine if edges are masked in charts.

Padding choices (zero, mirror, extend) change edges — document padding in code.

Daily majors workflow

1. Align business-day returns

Same snapshot discipline as other methods.

2. Mask edge cones in plots

Do not interpret bright R^2 at the very last dates without checking leakage.

3. Pre-specify wavelet, scales, smoothing

Scale grids tuned after seeing pair rankings invite overfit.

Versus spectral coherence

Method 21 (spectral coherence) assumes stationary spectra across time. Wavelets allow evolving linkage — useful around crises. Both are linear tools in their coefficient domain.

Neither replaces tail dependence counts (method 19) or DCCA long-memory scaling (method 23).

Software reproduction

```
import pywt
Wx, _ = pywt.cwt(x, scales, "morl")
Wy, _ = pywt.cwt(y, scales, "morl")
# correlate Wx[i] and Wy[i] per scale i
```

Full wavelet coherence toolkits implement smoothing S and significance contours — verify against Grinsted et al. when publishing figures.

Install PyWavelets in the environment that builds this PDF so asserts match runtime.

Limits

Scale labels are approximate in days — calibrate with known test signals. Complex-valued noise and volume gaps distort coefficients.

High ρ_wavelet(s) at one scale does not prove stable out-of-sample linkage.

Chasing max $R^2(s,t)$ across pairs without pre-registration overstates significance.

Practice

Q: Does R^2 show causation? **A:** No — phase hints at timing, not mechanism.

Q: Can edges be ignored? **A:** No — mask cone-of-influence regions.

Check: teaching $\rho_{wavelet}$ at scale 2.0 $\approx$ 0.9991.

Q: Is $\rho_{wavelet}$ the same as Pearson r? **A:** No — it is on wavelet coefficients at scale s.

Sources

[1] Grinsted, A., Moore, J. C., & Jevrejeva, S. (2004). Application of the cross wavelet transform and wavelet coherence to geophysical time series. Nonlinear Processes in Geophysics, 11, 561–566.
Cross wavelet and coherence reference.
Open source

[2] forex-formulas-explained.pdf — section 22 (this repository).
ρ_wavelet and R^2 formulas.
Open source

[3] PyWavelets documentation: pywt.cwt.
CWT implementation.
Open source

Rebuild: python build_guide.py. End of wavelet correlation and coherence beginner guide.